

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

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```

1. numpy digunakan untuk operasi numerik dan pengolahan array.
2. matplotlib.pyplot digunakan untuk membuat visualisasi atau grafik data.
3. pandas digunakan untuk membaca, mengelola, dan menganalisis data dalam bentuk tabel (DataFrame).

```
dataset = pd.read_csv('/content/Kabupatenn.csv', sep=';')
x = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
```

```
dataset = pd.read_csv('/content/Kabupaten.csv', sep=';') x = dataset.iloc[:, :-1].values y = dataset.iloc[:, -1].values
```

1. Membaca dataset Kabupaten.csv ke dalam program.
2. Variabel x berisi seluruh data fitur (No, Regency, Age, Salary).
3. Variabel y berisi data target yaitu Purchased.

```
print(x)
```

```
[[1 'Sleman' 25.0 4200000.0]
 [2 'Bantul' 31.0 nan]
 [3 'Kulon Progo' 47.0 8500000.0]
 [4 'Bantul' nan 5500000.0]
 [5 'Kulon Progo' 22.0 3800000.0]
 [6 'Sleman' 53.0 9200000.0]
 [7 'Bantul' 36.0 6000000.0]
 [8 'Sleman' nan 7100000.0]
 [9 'Kulon Progo' 28.0 nan]
 [10 'Sleman' 41.0 7800000.0]
 [11 'Bantul' nan 4600000.0]
 [12 'Sleman' 34.0 nan]
 [13 'Kulon Progo' 50.0 8900000.0]
 [14 'Bantul' 26.0 4000000.0]
 [15 'Sleman' nan 6300000.0]]
```

```
print(y)
```

```
['No' 'Yes' 'No' 'Yes' 'No' 'Yes' 'Yes' 'No' 'Yes' 'Yes' 'No' 'No' 'Yes'
 'No' 'Yes']
```

```
from sklearn.impute import SimpleImputer
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
x[:, 2:4] = imputer.fit_transform(x[:, 2:4])
```

```
from sklearn.impute import SimpleImputer imputer = SimpleImputer(missing_values=np.nan, strategy='mean') x[:, 2:4] =
imputer.fit_transform(x[:, 2:4])
```

1. Mengatasi data yang kosong (missing value) pada kolom Age dan Salary.
2. Nilai yang kosong diganti dengan rata-rata (mean) dari masing-masing kolom.
3. Bertujuan agar dataset menjadi lengkap dan dapat diproses lebih lanjut.

```
print(x)
```

```
[[1 'Sleman' 25.0 4200000.0]
 [2 'Bantul' 31.0 6325000.0]
 [3 'Kulon Progo' 47.0 8500000.0]
 [4 'Bantul' 35.72727272727273 5500000.0]
 [5 'Kulon Progo' 22.0 3800000.0]
 [6 'Sleman' 53.0 9200000.0]
 [7 'Bantul' 36.0 6000000.0]
 [8 'Sleman' 35.72727272727273 7100000.0]
 [9 'Kulon Progo' 28.0 6325000.0]
 [10 'Sleman' 41.0 7800000.0]
 [11 'Bantul' 35.72727272727273 4600000.0]
 [12 'Sleman' 34.0 6325000.0]
 [13 'Kulon Progo' 50.0 8900000.0]
 [14 'Bantul' 26.0 4000000.0]
 [15 'Sleman' 35.72727272727273 6300000.0]]
```

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
```

```
ct = ColumnTransformer(transformers=[('encoder', OneHotEncoder(), [1])],remainder='passthrough')
x = np.array(ct.fit_transform(x))
```

```
from sklearn.compose import ColumnTransformer from sklearn.preprocessing import OneHotEncoder ct =
ColumnTransformer(transformers=[('encoder', OneHotEncoder(), [1])],remainder='passthrough')x = np.array(ct.fit_transform(x))
```

1. Mengubah data teks pada kolom Regency menjadi data numerik.
2. Komputer tidak dapat mengolah data berupa teks secara langsung.
3. One Hot Encoding mengubah kategori seperti Sleman, Bantul, dan Kulon Progo menjadi bentuk angka 0 dan 1.

```
print(x)
```

```
[[0.0 0.0 1.0 1 25.0 4200000.0]
 [1.0 0.0 0.0 2 31.0 6325000.0]
 [0.0 1.0 0.0 3 47.0 8500000.0]
 [1.0 0.0 0.0 4 35.72727272727273 5500000.0]
 [0.0 1.0 0.0 5 22.0 3800000.0]
 [0.0 0.0 1.0 6 53.0 9200000.0]
 [1.0 0.0 0.0 7 36.0 6000000.0]
 [0.0 0.0 1.0 8 35.72727272727273 7100000.0]
 [0.0 1.0 0.0 9 28.0 6325000.0]
 [0.0 0.0 1.0 10 41.0 7800000.0]
 [1.0 0.0 0.0 11 35.72727272727273 4600000.0]
 [0.0 0.0 1.0 12 34.0 6325000.0]
 [0.0 1.0 0.0 13 50.0 8900000.0]
 [1.0 0.0 0.0 14 26.0 4000000.0]
 [0.0 0.0 1.0 15 35.72727272727273 6300000.0]]
```

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
y = le.fit_transform(y)
```

```
from sklearn.preprocessing import LabelEncoder le = LabelEncoder() y = le.fit_transform(y)
```

1. Mengubah data target Purchased dari bentuk teks menjadi angka.
2. Nilai "No" diubah menjadi 0.
3. Nilai "Yes" diubah menjadi 1.
4. Memudahkan proses analisis dan pemodelan data.

```
print(y)
```

```
[0 1 0 1 0 1 1 0 1 1 0 0 1 0 1]
```

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=1)
```

```
from sklearn.model_selection import train_test_split x_train, x_test, y_train, y_test = train_test_split( x, y, test_size=0.2, random_state=1)
```

1. Membagi dataset menjadi data latih (training) dan data uji (testing).
2. Sebanyak 80% data digunakan untuk pelatihan model.
3. Sebanyak 20% data digunakan untuk pengujian model.
4. Bertujuan untuk mengetahui performa model pada data baru.

```
print(x_train)
```

```
[[0.0 1.0 0.0 3 47.0 8500000.0]
 [1.0 0.0 0.0 11 35.72727272727273 4600000.0]
 [0.0 1.0 0.0 5 22.0 3800000.0]
 [1.0 0.0 0.0 2 31.0 6325000.0]
 [1.0 0.0 0.0 14 26.0 4000000.0]
 [0.0 0.0 1.0 1 25.0 4200000.0]
 [0.0 0.0 1.0 15 35.72727272727273 6300000.0]
 [0.0 0.0 1.0 10 41.0 7800000.0]
 [0.0 1.0 0.0 9 28.0 6325000.0]
 [0.0 1.0 0.0 13 50.0 8900000.0]
 [0.0 0.0 1.0 12 34.0 6325000.0]
 [0.0 0.0 1.0 6 53.0 9200000.0]]
```

```
print(x_test)
```

```
[[1.0 0.0 0.0 4 35.72727272727273 5500000.0]
 [0.0 0.0 1.0 8 35.72727272727273 7100000.0]
 [1.0 0.0 0.0 7 36.0 6000000.0]]
```

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
```

```
x_train[:, 4:] = sc.fit_transform(x_train[:, 4:])  
x_test[:, 4:] = sc.transform(x_test[:, 4:])
```

from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train[:, 4:] = sc.fit_transform(x_train[:, 4:])
x_test[:, 4:] = sc.transform(x_test[:, 4:])

1. Menyamakan skala antar fitur numerik.
2. Menghindari perbedaan nilai yang terlalu jauh antar variabel.
3. Membantu meningkatkan performa algoritma machine learning.
4. Data diubah sehingga memiliki rata-rata mendekati 0 dan standar deviasi mendekati 1.

```
print(x_train)
```

```
[[0.0 1.0 0.0 3 1.1588377289089473 1.158869760464728]  
 [1.0 0.0 0.0 11 0.0023316654505210867 -0.9493947600308704]  
 [0.0 1.0 0.0 5 -1.4059942666641752 -1.3818592770556084]  
 [1.0 0.0 0.0 2 -0.48265474825785104 -0.016893145196278832]  
 [1.0 0.0 0.0 14 -0.9956211473724755 -1.2737431477994239]  
 [0.0 0.0 1.0 1 -1.0982144271954004 -1.1656270185432394]  
 [0.0 0.0 1.0 15 0.0023316654505210867 -0.030407661353301897]  
 [0.0 0.0 1.0 10 0.5432780499713978 0.7804633080680821]  
 [0.0 1.0 0.0 9 -0.7904345877266258 -0.016893145196278832]  
 [0.0 1.0 0.0 13 1.4666175683777218 1.375102018977097]  
 [0.0 0.0 1.0 12 -0.17487490878907638 -0.016893145196278832]  
 [0.0 0.0 1.0 6 1.7743974078464966 1.5372762128613737]]
```

```
print(x_test)
```

```
[[1.0 0.0 0.0 4 0.0023316654505210867 -0.46287217837804]  
 [0.0 0.0 1.0 8 0.0023316654505210867 0.40205685567143623]  
 [1.0 0.0 0.0 7 0.030311650856773398 -0.1925818552375787]]
```